

Number and Algebra

Name: Mark Scheme

Total Marks: 32 marks

Total Time: 30 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Questions 1-5 are multiple choice one mark each.
Circle the correct answer.

[5 marks]

1. For the expression $3x^2 - 7x - 3y - 2$. The coefficient of x is

A 7 B 3 C -3

☒ D -7 E -2

2. Which of the following pairs are like terms?

A $10a$ and $-8a^2$ B $6m$ and mn C $8st^2$ and $-9s^2t$

D $2t$ and $2e$ ☒ E $4pq$ and $7pq$

3. Which of the following expressions can be simplified further by combining like terms?

A $8 - b + a$ B $9a + ab - 5$ ☒ C $6ab + 4b - ba$

D $7b - ba + 1$ E $8a + 2b - 6 + 9ab$

4. Which of the following is equivalent to $3(t + 2)$

A $3t$ B $3t + 2$ ☒ C $3t + 6$

D $6t$ E $3t + 5$

5. Which of the following is equivalent to $5a + 6 - 3a + 2$

A $2a + 4$ B $8a + 4$ C $10a$

☒ D $2a + 8$ E $8a + 8$

6. Simplify the following expressions

a. $2x + 5x$

$7x$ ✓

b. $3 \times a \times b$

$3ab$ ✓

c. $\frac{10p}{15}$

$\frac{2p}{3}$ ✓

d. $7t - 6t - 3$

$t - 3$ ✓

e. $\frac{9ab}{3b}$

$3a$ ✓

f. $2g \times 6h$

$12gh$ ✓

g. $9w - 4w^2 - 6w^2 - 2w$

$7w - 10w^2$ ✓✓

h. $5cd \times c^2e$

$5c^3de$ ✓✓

i. $12p^2q \div 15p$

$\frac{4pq}{5}$ ✓✓

[12 marks]

7. Expand and simplify if needed the following expressions

a. $3(c + 2)$

$3c + 6$ ✓

b. $-5(2c - 4)$

$-10c + 20$ ✓✓

c. $2x(5x - 4)$

$10x^2 - 8x$ ✓✓

d. $3(c + 2) - 5(2c - 4)$

$3c + 6 - 10c + 20$
 $= -7c + 26$
 $= 26 - 7c$ ✓✓

[7 marks]

8. Fully factorise the following Expressions

a. $4p + 12$

$4(p + 3)$ ✓✓

b. $10 - 15f$

$5(2 - 3f)$ ✓✓

c. $-2x - 6$

$-2(x + 3)$ ✓✓

d. $x^2 + 7x$

$x(x + 7)$ ✓✓

[8 marks]

